

Name _____

Math 1350, Quiz #4

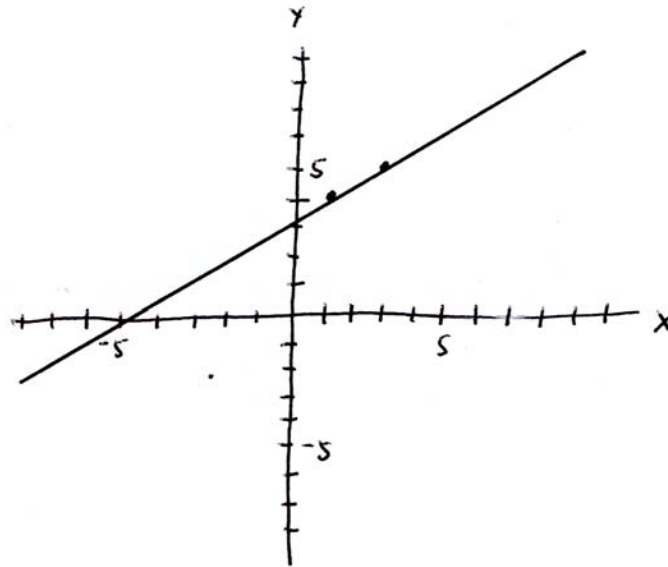
1. (Section 3.1, Problem 9) Find an equation of the line through the points (1,4) and (3,5) and sketch the graph.

$$m = \frac{5-4}{3-1} = \frac{1}{2}$$

$$y-4 = \frac{1}{2}(x-1)$$

or

$$y-5 = \frac{1}{2}(x-3)$$



2. (Section 3.2, Problem 21) Find an equation of the line parallel to $2x - 5y + 1 = 0$ and containing the point (2,3).

$$5y = 2x + 1$$

$$y = \frac{2}{5}x + \frac{1}{5}$$

$$m = \frac{2}{5}$$

$$y-3 = \frac{2}{5}(x-2)$$

3. (Section 4.1, Problem 17) Express the equation $x^2 + y^2 - 2x - 4y + 1 = 0$ in standard (center-radius) form and sketch the graph.

$$x^2 - 2x + \underline{1} + y^2 - 4y + \underline{4} = -1 + 1 + 4$$

$$(x-1)^2 + (y-2)^2 = 4$$

$$C(1, 2), r=2$$

